BACSE202 – DataBase Systems

Assessment 3 Lab

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Normalization of Event Management System: Blood Donation Camp

(A) Un-Normalized Form

Before normalization, all the information related to donors, events, staff, and donations was stored in a single table. This resulted in repeated values and unnecessary redundancy.

Donation_ID	Donor_ID	Donor_Name	Age	Gender	Blood_Group	Phone_No	Event_ID
401	201	Rahul Sharma	22	Male	B+	9876543210	101
402	201	Rahul Sharma	22	Male	B+	9876543210	102
403	202	Priya Nair	20	Female	O+	9876501234	101

Observation

From the above table it can be observed that:

- Donor information is repeated for every donation made by the same donor.
- Event details are repeated for every donor participating in that event.
- Staff details are repeated whenever the same staff member supervises multiple donations.

Therefore, the database contains data redundancy.

(B) Functional Dependencies (FD)

The primary key of the unnormalized table is Donation_ID, which uniquely identifies every donation record.

Functional Dependency

Donation_ID → Donor_ID,Donor_Name,Age,Gender,Blood_Group

Phone_No,Event_ID,Event_Name,Event_Date,
Venue,Staff_ID,Staff_Name,Role,Quantity.

(C) Anomalies

Update Anomaly

Changing the venue of Event 101 requires updating multiple rows. Missing one update causes inconsistent data.

Insertion Anomaly

A new event cannot be inserted unless donor details are also entered.

Deletion Anomaly

Deleting the only donation for an event removes the event and staff information as well.

(D) First Normal Form (1NF)

Donation_ID	Donor_ID	Donor_Name	Age	Gender	Blood_Group	Phone_No	Event_ID
401	201	Rahul Sharma	22	Male	B+	9876543210	101
402	201	Rahul Sharma	22	Male	B+	9876543210	102
403	202	Priya Nair	20	Female	O+	9876501234	101

Observation

The UNF table already contains atomic values. Therefore, no attribute contains multiple values in a single cell.

The table satisfies the conditions of First Normal Form.

(E) Second Normal Form (2NF)

(i) Donor Table

Donor_ID	Donor_Name	Age	Gender	Blood_Group	Phone_No
201	Rahul Sharma	22	Male	B+	9876543210
202	Priya Nair	20	Female	O+	9876501234

(ii) Event Table

Event_ID	Event_Name	Event_Date	Venue
101	VIT Blood Donation Camp	15-03-2026	Anna Auditorium
102	NSS Blood Donation Camp	10-04-2026	VIT Hall

(iii) Donation Table

Donation_ID	Donor_ID	Event_ID	Quantity
401	201	101	450
402	201	102	450
403	202	101	450

Observation

The relation is decomposed into DONOR, EVENT and DONATION to remove redundancy and partial dependencies. Details of individual tables are only stored once.

Therefore, the database satisfies the conditions of Second Normal Form (2NF).

Functional Dependency

(i) Donor_ID → Donor_Name, Age, Gender, Blood_Group, Phone_No

(ii) Event_ID → Event_Name, Event_Date, Venue

(iii) Donation_ID → Donor_ID, Event_ID, Quantity

(F) Third Normal Form (3NF)

(i) Donor Table

Donor_ID (PK)	Donor_Name	Age	Gender	Blood_Group	Phone_No	Email	Address
201	Rahul Sharma	22	Male	B+	9876543210	rahul@gmail.com	Chennai
202	Priya Nair	20	Female	O+	9876501234	priya@gmail.com	Hyderabad

(ii) Event Table

Event_ID (PK)	Event_Name	Event_Date	Venue	Organizer	Start_Time	End_Time	Status
101	VIT Blood Donation Camp	15-03-2026	Anna Auditorium	Red Cross Society	09:00	16:00	Scheduled
102	NSS Blood Donation Camp	10-04-2026	VIT Hall	NSS Club	09:30	15:00	Scheduled

(iii) Donation Table

Donation_ID (PK)	Donation_Date	Donation_Time	Quantity	Blood_Pressure	Hemoglobin
401	15-03-2026	10:30	450	120/80	14.2

(iv) Staff Table

Staff_ID (PK)	Staff_Name	Role	Phone_No	Email	Event_ID (FK)
301	Dr. Anil Kumar	Medical Officer	9123456789	anil@vit.ac.in	101
302	Dr. Meera	Doctor	9876123456	meera@vit.ac.in	102

Observation

After removing transitive dependencies, the database is decomposed into four relations: EVENT, DONOR, STAFF and DONATION. Donation details reference donors and events using foreign keys.

There are no partial dependencies and no transitive dependencies.

Hence, the database satisfies Third Normal Form (3NF).

Functional Dependency

(i) Donor_ID → Donor_Name, Age, Gender, Blood_Group, Phone_No, Email, Address, Weight_kg

(ii) Event_ID → Event_Name, Event_Date, Venue, Organizer, Start_Time, End_Time, Status

(iii) Donation_ID → Donation_Date, Donation_Time, Quantity, Blood_Pressure, Hemoglobin, Eligibility_Status, HIV_Result, Donor_ID, Event_ID

(iv) Staff_ID → Staff_Name, Role, Phone_No, Email, Event_ID

(G) Final Relational Schema

EVENT(Event_ID (PK),Event_Name,Event_Date,
Venue,Organizer,Start_Time,End_Time,
Status)

DONOR(Donor_ID (PK),Donor_Name,Age,Gender,
Blood_Group,Phone_No,Email,Address,
Weight_kg)

STAFF(Staff_ID (PK),Staff_Name,Role,

Phone_No,Email,Event_ID (FK))

DONATION(Donation_ID (PK),Donation_Date,
Donation_Time,Quantity,Blood_Pressure,
Hemoglobin,Eligibility_Status,HIV_Result,
Donor_ID (FK),Event_ID (FK))

(H) mySQL Codes

1NF

```
CREATE TABLE BLOOD_DONATION_1NF (
    Donation_ID INT PRIMARY KEY,
    Donor_ID INT,
    Donor_Name VARCHAR(100),
    Age INT,
    Gender VARCHAR(10),
    Blood_Group VARCHAR(5),
    Phone_No VARCHAR(15),
    Event_ID INT,
    Event_Name VARCHAR(100),
    Event_Date DATE,
    Venue VARCHAR(100),
    Staff_ID INT,
    Staff_Name VARCHAR(100),
    Role VARCHAR(50),
    Quantity INT
);
```

Donation_ID	Donor_Name	Event_Name	Staff_Name	Blood_Group	Quantity
401	Rahul Sharma	VIT Camp	Dr. Anil	B+	450
402	Rahul Sharma	NSS Camp	Dr. Meera	B+	450

2NF

```
CREATE TABLE DONOR_2NF (
 Donor_ID INT PRIMARY KEY,
 Donor_Name VARCHAR(100),
 Age INT,
 Gender VARCHAR(10),
 Blood_Group VARCHAR(5),
 Phone_No VARCHAR(15)
);

CREATE TABLE EVENT_2NF (
 Event_ID INT PRIMARY KEY,
 Event_Name VARCHAR(100),
 Event_Date DATE,
 Venue VARCHAR(100)
);

CREATE TABLE DONATION_2NF (
 Donation_ID INT PRIMARY KEY,
 Donor_ID INT,
 Event_ID INT,
 Quantity INT,
```

```
 FOREIGN KEY (Donor_ID) REFERENCES DONOR_2NF(Donor_ID),
 FOREIGN KEY (Event_ID) REFERENCES EVENT_2NF(Event_ID)
);
```

DONOR_2NF

Donor_ID	Donor_Name	Blood_Group
201	Rahul Sharma	B+
202	Priya Nair	O+

EVENT_2NF

Event_ID	Event_Name	Venue
101	VIT Camp	Anna Auditorium
102	NSS Camp	VIT Hall

DONATION_2NF

Donation_ID	Donor_ID	Event_ID	Quantity
401	201	101	450
402	202	102	450

3NF

```
CREATE TABLE EVENT (
 Event_ID INT PRIMARY KEY,
 Event_Name VARCHAR(100),
 Event_Date DATE,
 Venue VARCHAR(100),
 Organizer VARCHAR(100),
 Status VARCHAR(20)
);

CREATE TABLE DONOR (
 Donor_ID INT PRIMARY KEY,
 Donor_Name VARCHAR(100),
 Age INT,
 Gender VARCHAR(10),
 Blood_Group VARCHAR(5),
 Phone_No VARCHAR(15),
 Email VARCHAR(100),
 Address VARCHAR(100),
 Weight_kg DECIMAL(5,2)
);

CREATE TABLE STAFF (
 Staff_ID INT PRIMARY KEY,
 Staff_Name VARCHAR(100),
 Role VARCHAR(50),
 Event_ID INT,
 FOREIGN KEY (Event_ID) REFERENCES EVENT(Event_ID)
);

CREATE TABLE DONATION (
 Donation_ID INT PRIMARY KEY,
 Donation_Date DATE,
```

```
 Quantity INT,
 Donor_ID INT,
 Event_ID INT,
 FOREIGN KEY (Donor_ID) REFERENCES DONOR(Donor_ID),
 FOREIGN KEY (Event_ID) REFERENCES EVENT(Event_ID)
);
```

Event_ID	Event_Name	Venue
101	VIT Camp	Anna Auditorium

DONOR

Donor_ID	Donor_Name	Blood_Group
201	Rahul Sharma	B+

STAFF

Staff_ID	Staff_Name	Role
301	Dr. Anil	Medical Officer

DONATION

Donation_ID	Donor_ID	Event_ID	Quantity
401	201	101	450

(I) Advantages of Normalization

- Eliminates duplicate data.
- Reduces data redundancy.
- Improves data consistency.
- Maintains data integrity.
- Simplifies database maintenance.
- Improves scalability and overall database performance.

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